

//SIT 55 : ASSIGNMENT 5 : INTERFACE

/*Design and develop a context for given case study and implement an interface for Vehicles . Consider the example of vehicles like bicycle, car and bike.All Vehicles have common functionalities such as Gear C change, Speed up and apply breaks. Make an interface and put all these common functionalities. Bicycle, Bike, Car classes should be implemented for all these functionalities in their own class in their own way.*/

```
interface Vehicles
{
    void change_gear();
    void speed();
    void breaks();
}
class Car implements Vehicles
{
    public void change_gear()
    {
        System.out.println("CAR HAS 5 GEARS");
    }
    public void speed()
    {
        System.out.println("MAX.SPEED OF CAR IS 400 KPH");
    }
    public void breaks()
    {
        System.out.println("CAR HAS GOOD BREAKS");
    }
}
class Bike implements Vehicles
{
    public void change_gear()
    {
        System.out.println("BIKE HAS 5 GEARS");
    }
    public void speed()
    {
        System.out.println("MAX.SPEED OF CAR IS 150 KPH");
    }
    public void breaks()
    {
        System.out.println("BIKE HAS GOOD BREAKS");
    }
}
class Cycle implements Vehicles
{
    public void change_gear()
    {
        System.out.println("CYCLE HAS NO GEARS");
    }
    public void speed()
    {
        System.out.println("MAX.SPEED OF CAR IS 22 KPH");
    }
    public void breaks()
    {

```

```

    System.out.println("CYCLE HAS GOOD BREAKS");
}
}
public class AssignInterface
{
    public static void main(String args[])
    {
        Car c=new Car();
        Bike b=new Bike();
        Cycle bc=new Cycle();

        c.change_gear();
        c.speed();
        c.breaks();
        System.out.println();
        b.change_gear();
        b.speed();
        b.breaks();
        System.out.println();
        bc.change_gear();
        bc.speed();
        bc.breaks();
    }
}

```

/*OUTPUT:

CAR HAS 5 GEARS
 MAX.SPEED OF CAR IS 400 KPH
 CAR HAS GOOD BREAKS

BIKE HAS 5 GEARS
 MAX.SPEED OF CAR IS 150 KPH
 BIKE HAS GOOD BREAKS

CYCLE HAS NO GEARS
 MAX.SPEED OF CAR IS 22 KPH
 CYCLE HAS GOOD BREAKS*/